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Institutional Corporate Bond Pricing

Contents

1 Big picture

- The paper builds and estimates an **equilibrium corporate bond pricing model** where **prices clear markets given institutional demand** (rather than a representative-agent SDF-only view).
- Key state variable: **composition of institutional demand** (who holds the bonds) matters for equilibrium credit spreads and corporate funding costs.
- Core empirical fact motivating the model: strong **segmentation across rating groups** (investment grade vs high yield) and strong **heterogeneity across institution types** (insurers vs mutual funds vs ETFs, etc.).
- Estimated elasticities are **finite and economically small**: market-wide elasticity around ≈ 6, with mutual funds much more elastic, and life insurers/ETFs very inelastic.

2 Data

- Holdings: Thomson Reuters eMAXX (quarterly, security-level).
- Prices/yields/ratings: WRDS Bond Returns (TRACE-based).
- Characteristics: FISD.
- Sample: 2006Q1–2020Q3; institutions in sample hold roughly 45–50% of total corporate bonds outstanding (in par/value terms as described).
- There is large segmentation IG vs HY: many institutions invest almost exclusively in one segment (especially insurers and ETFs).

3 Models

3.1 Notation and portfolio weight decomposition

Investors $i = 1, \dots, I$. Bonds $n = 0, \dots, N$, where $n = 0$ is the outside asset. Time t. Yield of bond n at time t: $y_t(n)$. Bond characteristics: $x_t(n)$ (maturity, rating, issuance size, bid-ask, plus controls like coupon in estimation).

Investor wealth (AUM): $A_{i,t}$.

Bonds belong to rating group $l \in \{1, 2\}$: $l = 1$ (IG), $l = 2$ (HY). Portfolio weight decomposition:

$$w_{i,t}(n) = w_{i,t}(n, l) = w_{i,t}(n \mid l)\, w_{i,t}(l). \tag{1}$$

Within-group and across-group weights satisfy:

$$\sum_{n=0}^{N} w_{i,t}(n \mid l) = 1, \tag{2}$$

$$w_{i,t}(1) + w_{i,t}(2) = 1. \tag{3}$$

3.2 Within-rating-group demand (logit within nests)

Within rating group l, the portfolio weight is:

$$w_{i,t}(n \mid l) = \frac{\delta_{i,t}(n,l)}{1 + \sum_{m=1}^{N} \delta_{i,t}(m,l)}, \tag{4}$$

$$\delta_{i,t}(n,l) = \exp\Big(\beta_{0,i,l}\, y_t(n) + \beta_{1,i,l}' x_t(n) + \epsilon_{i,t}(n)\Big). \tag{5}$$

Outside-asset share within group l:

$$w_{i,t}(0 \mid l) = \frac{1}{1 + \sum_{m=1}^{N} \delta_{i,t}(m,l)}. \tag{6}$$

Log-odds form (estimating equation for within-group demand):

$$\log\left(\frac{w_{i,t}(n \mid l)}{w_{i,t}(0 \mid l)}\right) = \beta_{0,i,l}\, y_t(n) + \beta_{1,i,l}' x_t(n) + \epsilon_{i,t}(n). \tag{7}$$

My interpretation of parameters (within nests).

- $\beta_{0,i,l}$: sensitivity of demand to yields (hence, inverse sensitivity to prices). Larger $\beta_{0,i,l}$ implies more elastic behavior within that rating group.
- $\beta_{1,i,l}$: preference loadings for bond characteristics (maturity/duration, liquidity, size, credit risk proxies).
- $\epsilon_{i,t}(n)$: latent (unobserved) demand component.

3.3 Across-rating-group allocation (nested logit across nests)

Across-group weights:

$$w_{i,t}(l) = \frac{\Big(1 + \sum_{m=0}^{N} \delta_{i,t}(m,l)\Big)^{\lambda_{i,l}} \exp\left(\alpha_l + \xi_{i,t}(l)\right)}{\sum_{k=1}^{2} \Big(1 + \sum_{m=0}^{N} \delta_{i,t}(m,k)\Big)^{\lambda_{i,k}} \exp\left(\alpha_k + \xi_{i,t}(k)\right)}. \tag{8}$$

Taking the ratio IG vs HY and logs gives:

$$\log\left(\frac{w_{i,t}(1)}{w_{i,t}(2)}\right) = \lambda_{i,1} \log\left(1 + \sum_{m=1}^{N} \delta_{i,t}(m,1)\right) - \lambda_{i,2} \log\left(1 + \sum_{m=1}^{N} \delta_{i,t}(m,2)\right) + \alpha_1 + \xi_{i,t}(1) \tag{9}$$

$$= -\lambda_{i,1} \log\left(w_{i,t}(0 \mid 1)\right) + \lambda_{i,2} \log\left(w_{i,t}(0 \mid 2)\right) + \alpha_1 + \xi_{i,t}(1). \tag{10}$$

Normalization: $\alpha_2 + \xi_{i,t}(2) = 0$ for HY.

My interpretation of $\lambda_{i,l}$ (segmentation).

- $\lambda_{i,l} \in [0,1]$ controls **how much investor i reallocates across IG vs HY when a whole group becomes more/less attractive** (via the "inclusive value").
- $\lambda_{i,l} = 0$: near **full segmentation** across groups (allocations across rating groups barely respond to relative attractiveness).
- $\lambda_{i,l} = 1$: **no extra segmentation layer** (collapses to one-tier logit).

3.4 Special case: nonnested model ($\lambda_{i,1} = \lambda_{i,2} = 1$)

Then:

$$w_{i,t}(n) = w_{i,t}(n,l) = \frac{\delta_{i,t}(n,l)}{2 + \sum_{k=1}^{2} \sum_{m=1}^{N} \delta_{i,t}(m,k)}. \tag{11}$$

This corresponds to the single-tier demand system used in related demand-system asset pricing work.

3.5 Market clearing (equilibrium pricing condition)

For each bond n:

$$M_t(n) = \sum_{i=1}^{I} A_{i,t}\, w_{i,t}(n), \tag{12}$$

where $M_t(n)$ is market value of bond n. (The paper also defines a residual sector for holders outside the eMAXX sample and very small institutions.)

3.6 Identification and instruments (sequential IV strategy)

Main issue: yields $y_t(n)$ are endogenous because latent demand $\epsilon_{i,t}(n)$ can move prices/yields.

They construct an instrument based on (inferred) **investment mandates / investment universes** (motivated by strong persistence in holdings). Instrument for yields (mandate-based demand pressure):

$$\tilde{y}_{i,t}(n) = \log\left(\sum_{j \neq i} A_{j,t} \frac{\mathbf{1}_{j,t}(n)}{1 + \sum_{m=1}^{N} \mathbf{1}_{j,t}(m)}\right), \tag{13}$$

where $\mathbf{1}_{j,t}(n) = 1$ if bond n belongs to investor j's mandate set.

Sequential estimation logic I note.

1. IV-estimate within-group equation (7) using $\tilde{y}_{i,t}(n)$ for yields.
2. Use predicted $\hat{\delta}_{i,t}(n,l)$ to compute $\hat{w}_{i,t}(0 \mid l)$ and instrument the across-group regressors in (10).
3. Estimate (10) with instruments $\log(\hat{w}_{i,t}(0 \mid 1))$ and $\log(\hat{w}_{i,t}(0 \mid 2))$.

3.7 Demand elasticities (investor-level, type-level, market-wide)

Define log demand of investor i for bond n in group l:

$$q_{i,t}(n,l) = \log\Big(A_{i,t} w_{i,t}(l) w_{i,t}(n \mid l)\Big) - p_t(n,l), \tag{14}$$

where $p_t(n,l)$ is log price of the bond.

Investor-bond own-price elasticity:

$$-\frac{\partial q_{i,t}(n,l)}{\partial p_t(n,l)} = 1 - \big(1 - w_{i,t}(l)\big)\, w_{i,t}(n \mid l)\, \lambda_{i,l}\, \beta^{p}_{0,i,l} - \big(1 - w_{i,t}(n \mid l)\big)\, \beta^{p}_{0,i,l}. \tag{15}$$

($\beta^p_{0,i,l}$ is the price-elasticity mapping of the yield coefficient, i.e., yield sensitivity scaled by maturity in the paper's discussion.)

Type-level elasticity (value-weighted within type):

$$-\frac{\partial q_{t,\text{type}}(n,l)}{\partial p_t(n,l)} = \sum_{i:\mathbf{1}_{\text{type}}(i)=1} \left(\frac{A_{i,t} w_{i,t}(n,l)}{\sum_{j:\mathbf{1}_{\text{type}}(j)=1} A_{j,t} w_{j,t}(n,l)} \right) \left(-\frac{\partial q_{i,t}(n,l)}{\partial p_t(n,l)} \right). \tag{16}$$

Market-wide demand and elasticity:

$$q_t(n,l) = \log \left(\sum_i A_{i,t} w_{i,t}(l) w_{i,t}(n \mid l) \right) - p_t(n,l), \tag{17}$$

$$-\frac{\partial q_t(n,l)}{\partial p_t(n,l)} = \sum_i \left(\frac{A_{i,t} w_{i,t}(n,l)}{\sum_j A_{j,t} w_{j,t}(n,l)} \right) \left(-\frac{\partial q_{i,t}(n,l)}{\partial p_t(n,l)} \right). \tag{18}$$

(They also define cross-elasticities within and across rating groups; these fill out the aggregate elasticity matrix $-\partial q_t / \partial p_t'$.)

3.8 Price impact / coliquidity (shock-to-price mapping)

Price impact of an idiosyncratic latent demand shock:

$$\frac{\partial p_t(n,l)}{\partial \epsilon_{i,t}(n,l)} = \left[-\frac{\partial q_t}{\partial p_t'} \right]^{-1} \cdot \frac{A_{i,t} w_{i,t}(n,l) \left(1 - w_{i,t}(n \mid l) + \lambda_{i,l} w_{i,t}(n \mid l)(1 - w_{i,t}(l))\right)}{\sum_j A_{j,t} w_{j,t}(n,l)}. \tag{19}$$

To decompose heterogeneity in price impact, they average demand slopes within type:

$$\bar{\beta}^p_{0,\text{type},l} = \frac{\sum_{i:\mathbf{1}_{\text{type}}(i)=1} A_{i,t} w_{i,t}(n,l)\, \hat{\beta}^p_{0,i,l}}{\sum_{j:\mathbf{1}_{\text{type}}(j)=1} A_{j,t} w_{j,t}(n,l)}. \tag{20}$$

3.9 Counterfactual equilibrium mapping

They summarize equilibrium prices as:

$$p_t = g(s_t, x_t, A_t, \beta_t, \lambda, \epsilon_t), \tag{21}$$

and compute counterfactual price changes such as:

$$\Delta p_t = g(s_t, x_t, A_t^{CF}, \beta_t, \lambda, \epsilon_t) - g(s_t, x_t, A_t, \beta_t, \lambda, \epsilon_t). \tag{22}$$

(They then translate price changes into spread changes using their pseudo-zero-coupon yield construction.)

4 Tables

4.1 Table 1: Summary of institutional holdings (sample size and representativeness)

Table 1: Summary of institutional holdings (Table 1)

Year	# Funds	% Market held	Fund AUM (USD mm)		# Bonds held	
			Median	90th pct	Median	90th pct
2006	1,281	49	54	629	48	162
2007	1,360	45	55	623	51	168
2008	1,570	45	55	618	53	182
2009	1,972	46	59	639	57	212
2010	2,036	50	63	726	58	216
2011	2,172	48	65	757	60	229
2012	2,444	49	68	770	64	236
2013	2,486	48	71	831	68	252
2014	2,622	47	70	853	67	258
2015	2,676	46	70	872	69	278
2016	3,260	45	67	792	68	282
2017	3,666	48	69	848	74	305
2018	3,297	45	72	879	79	331
2019	3,960	45	68	806	78	328
2020	3,478	44	76	983	86	377

Notes: Each cell is the time-series mean of the quarterly summary statistic within the year.
Sample: 2006Q1–2020Q3.

My interpretation.

- Coverage: institutions in the sample consistently hold $\approx 44\% - 50\%$ of the corporate bond market outstanding (large enough to study equilibrium pricing with a residual sector).
- The number of funds grows substantially over time (roughly tripling from 2006 to the peak years), consistent with broader institutional growth / granularity.
- Median AUM and the 90th percentile AUM increase materially over time; portfolios also become broader (median number of bonds held rises).

4.2 Table 2: Persistence of the set of bonds held (mandates motivation)

Table 2: Persistence of the set of bonds held (Table 2)

AUM pctile	% of currently held bonds also held in any of the previous k quarters										
	1	2	3	4	5	6	7	8	9	10	11
1	92	95	95	96	97	97	97	98	98	98	98
2	91	94	95	96	96	97	97	98	98	98	98
3	91	93	94	95	96	96	97	97	97	98	98
4	91	94	95	95	96	96	97	97	97	98	98
5	91	94	95	95	96	96	97	97	97	98	98
6	91	94	95	96	96	96	97	97	97	98	98
7	91	94	95	96	96	97	97	97	98	98	98
8	91	94	95	96	96	97	97	97	98	98	98
9	91	95	96	96	97	97	97	98	98	98	98
10	91	95	97	97	98	98	98	98	98	98	99

Notes: Pooled median across time and institutions in each AUM percentile. Sample: 2006Q1–2020Q3.

My interpretation.

- Portfolios are **extremely persistent**: even at $k = 1$ quarter, around 91–92% of positions are from the recent set, and by 11 quarters it rises to $\approx 98\% - 99\%$.
- This supports the **investment-universe / mandate** idea used for the yield instrument: institutions behave like they choose within a stable set rather than the whole market every quarter.

4.3 Table 3: First-stage t-statistics (instrument strength)

Table 3: First-stage t-statistics for instruments (Table 3)

Panel A: pooled first-stage t-statistics by institution type

	Bond yields	$\log\big(\hat{w}_{i,t}(0 \mid 1)\big)$	$\log\big(\hat{w}_{i,t}(0 \mid 2)\big)$
Life insurers	22.3	76.5	194.7
P&C insurers	19.5	107.5	174.7
Mutual funds	17.8	103.9	192.0
Variable annuities	20.3	79.0	142.2
ETFs & index	13.5	17.3	45.8
Others	9.7	34.8	54.3
Foreign investors	15.8	68.1	131.6

Panel B: median first-stage t-statistics for the bond-yield instrument (quarter-by-quarter regressions)

Institution type	Median t	Time period	Median t
Life insurers	17.2	2006–2008	11.3
P&C insurers	11.3	2009–2010	9.8
Mutual funds	8.7	2011–2012	11.2
Variable annuities	7.4	2013–2014	16.0
ETFs & index	15.1	2015–2016	8.2
Others	7.7	2017–2018	8.7
Foreign investors	5.4	2019–2020	7.6

Notes: Panel A reports pooled first-stage t-statistics for the three instruments. Panel B reports median t-stats from institution-type-by-quarter first stages for the bond-yield instrument.

My interpretation.

- The mandate-based IVs are **very strong**: pooled t-stats are mostly $\gg 10$, with particularly large values for the outside-asset instruments.
- Even in quarter-by-quarter regressions (Panel B), median t-stats remain comfortably above typical weak-IV thresholds, suggesting the approach is statistically well-powered.

4.4 Table 4: Special case (nonnested) demand parameters by institution type

Table 4: Special case ($\lambda_1 = \lambda_2 = 1$): Heterogeneity in demand parameters (Table 4)

	Insurance		Asset management firms			Others	
	Life	P&C	Mutual funds	Variable annuity	ETFs & index	Others	Foreign
$\text{Yield}_{n,t}$	-0.131^{**}	0.146	0.826^{***}	0.608^{***}	-0.058	0.470^{**}	0.279^{***}
	(0.064)	(0.123)	(0.119)	(0.066)	(0.068)	(0.198)	(0.054)
$\text{Maturity}_{n,t}$	0.061^{**}	-0.046	-0.135^{***}	-0.125^{***}	0.042^{*}	-0.097^{*}	-0.018^{*}
	(0.026)	(0.029)	(0.028)	(0.016)	(0.023)	(0.058)	(0.009)
$\text{Bid-ask}_{n,t}$	0.017^{*}	-0.050	-0.215^{***}	-0.158^{***}	-0.009	-0.083^{**}	-0.114^{***}
	(0.010)	(0.036)	(0.024)	(0.021)	(0.011)	(0.033)	(0.019)
$\text{Issuance size}_{n,t}$	0.079^{***}	0.056^{***}	0.144^{***}	0.136^{***}	0.398^{***}	0.082^{***}	0.159^{***}
	(0.013)	(0.010)	(0.008)	(0.010)	(0.007)	(0.013)	(0.014)
$\text{Rating}_{n,t}$	-0.049^{*}	-0.221^{***}	-0.389^{***}	-0.354^{***}	0.025	-0.272^{***}	-0.147^{***}
	(0.027)	(0.049)	(0.045)	(0.038)	(0.019)	(0.053)	(0.040)
Fund × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,873,213	3,314,273	3,460,156	1,144,847	1,793,730	364,796	1,754,728

Notes: Dependent variable is $\log(w_{i,t}(n)/w_{i,t}(0))$. Yield is instrumented. Variables standardized (divided by pooled SD). Standard errors clustered at fund level. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

My interpretation.

- **Elasticity differences are huge.** Mutual funds and VA funds load strongly positively on yields (high yield sensitivity ⇒ more elastic demand). Life insurers are much less elastic and even show a negative yield coefficient here.
- **Preferred habitats / mandates show up in characteristics:**
  - Life insurers: positive maturity coefficient (tilt to longer maturity).
  - Mutual funds/VA: negative maturity coefficient (tilt to shorter maturity).
- **Liquidity:** mutual funds strongly avoid high bid-ask (prefer liquid bonds). Life insurers slightly prefer illiquidity (positive bid-ask coefficient).
- **Size:** ETFs have an extremely strong preference for larger issuance size (consistent with capitalization-weighted bond indices and size screens).
- **Credit risk:** ratings coefficients are negative for most types (conditional on yields, safer bonds are preferred); the ETF coefficient is near zero here (consistent with benchmark-like behavior rather than fine credit sorting in this reduced form).

4.5 Table 5: Nested model within rating groups (IG vs HY), pooled

Table 5: Nested model: within-rating-group demand estimates (Table 5)

	Life		P&C		Mutual		VA		ETF/Index		Others		Foreign	
	IG	HY	IG	HY	IG	HY	IG	HY	IG	HY	IG	HY	IG	HY
$\text{Yield}_{n,t}$	-0.031	0.018	0.356^{***}	0.211^{***}	0.965^{***}	0.324^{***}	0.621^{***}	0.403^{***}	0.042	0.056	0.571^{***}	0.337^{***}	0.160^{***}	0.200^{***}
	(0.064)	(0.126)	(0.121)	(0.073)	(0.143)	(0.055)	(0.065)	(0.066)	(0.142)	(0.117)	(0.140)	(0.127)	(0.038)	(0.064)
$\text{Maturity}_{n,t}$	0.029	-0.014	-0.088^{***}	-0.004	-0.152^{***}	-0.050^{***}	-0.128^{***}	-0.075^{***}	-0.012	0.025	-0.124^{***}	-0.139^{***}	-0.019^{***}	0.005
	(0.025)	(0.042)	(0.032)	(0.042)	(0.031)	(0.013)	(0.013)	(0.013)	(0.056)	(0.047)	(0.043)	(0.034)	(0.006)	(0.012)
$\text{Bid-ask}_{n,t}$	-0.004	-0.016	-0.103^{***}	-0.120^{***}	-0.245^{***}	-0.110^{***}	-0.157^{***}	-0.131^{***}	-0.026	-0.085^{***}	-0.105^{***}	-0.065^{***}	-0.065^{***}	-0.103^{***}
	(0.010)	(0.021)	(0.033)	(0.017)	(0.032)	(0.014)	(0.016)	(0.013)	(0.019)	(0.017)	(0.023)	(0.018)	(0.013)	(0.017)
$\text{Issuance size}_{n,t}$	0.093^{***}	0.062^{*}	0.053^{***}	0.065^{**}	0.133^{***}	0.159^{***}	0.132^{***}	0.133^{***}	0.405^{***}	0.364^{***}	0.092^{***}	0.081^{***}	0.165^{***}	0.142^{***}
	(0.013)	(0.032)	(0.008)	(0.027)	(0.011)	(0.011)	(0.009)	(0.012)	(0.005)	(0.010)	(0.011)	(0.020)	(0.017)	(0.018)
$\text{Rating}_{n,t}$	-0.003	-0.102^{*}	-0.234^{***}	-0.137^{***}	-0.420^{***}	-0.124^{***}	-0.278^{***}	-0.169^{***}	0.017	-0.080^{**}	-0.205^{***}	-0.169^{***}	-0.073^{***}	-0.079^{***}
	(0.027)	(0.059)	(0.052)	(0.051)	(0.045)	(0.033)	(0.036)	(0.035)	(0.037)	(0.038)	(0.056)	(0.055)	(0.018)	(0.029)
Fund × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,048,881	759,871	2,950,669	329,036	2,196,093	1,183,631	749,083	371,183	1,677,100	108,381	325,237	34,134	1,063,926	656,472

Notes: Within-group demand equation; dependent variable is $\log(w_{i,t}(n \mid l)/w_{i,t}(0 \mid l))$. Variables standardized. Yield instrumented. Fund-level clustered SEs. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

My interpretation.

- **Elasticities differ by rating group:** for many types (notably mutual funds), IG yield coefficients are larger than HY yield coefficients, consistent with more substitution within IG and/or stronger trading ability in IG.
- **Life insurers:** within IG, yield and rating sensitivity are close to zero; they appear to mainly care about the broad IG/HY split rather than fine IG credit distinctions.
- **Mutual funds:** strongly positive yield sensitivity (especially IG), and strongly negative rating coefficients even within groups (they tilt toward safer bonds within IG and within HY).
- **ETFs:** very large issuance-size loadings (index/benchmark footprint); modest yield sensitivity; meaningful liquidity aversion in HY (negative bid-ask).

4.6 Table 6: Nested model across rating groups (segmentation parameters λ)

Table 6: Nested model: Across rating group demand estimates (Table 6)

	Life	P&C	Mutual funds	VA	ETF/Index	Others	Foreign
Investment grade	0.193	0.269^{***}	0.176^{***}	0.252^{***}	0.108	0.114	0.234^{***}
	(0.136)	(0.088)	(0.033)	(0.044)	(0.087)	(0.073)	(0.043)
High yield	0.020	0.108^{***}	0.119^{***}	0.084^{***}	0.197	0.027	0.069^{*}
	(0.017)	(0.030)	(0.025)	(0.021)	(0.276)	(0.046)	(0.036)
Observations	9,252	4,835	13,260	4,246	348	515	5,761

Notes: Across-group regression of $\log(w_{i,t}(1)/w_{i,t}(2))$ on $\log(\hat{w}_{i,t}(0 \mid 1))$ and $\log(\hat{w}_{i,t}(0 \mid 2))$. SEs clustered at fund level. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

My interpretation.

- λ's are **well below 1** (often around 0.1–0.27): this is quantitative evidence of **imperfect substitution across IG and HY** (segmented market structure).

- The values are not near zero either ⇒ segmentation is **partial**: there is still cross-group reallocation when inclusive values move.
- Mutual funds and VA funds have precisely estimated, low λ, consistent with mandates and/or strong specialization patterns across the IG/HY boundary.

4.7 Table 7: Demand elasticities (type-level and market-wide)

Table 7: Demand elasticities (Table 7)

	Full sample			2006Q1–2009Q4			2010Q1–2020Q3		
	All	IG	HY	All	IG	HY	All	IG	HY
Life insurers	1.04	0.98	2.59	2.69	2.57	4.26	0.42	0.48	2.08
P&C insurers	7.60	7.70	5.43	4.97	5.02	5.15	8.58	8.60	5.61
Mutual funds	14.25	18.22	6.31	12.39	16.65	6.21	14.94	19.02	6.33
Variable annuities	8.43	9.93	6.09	6.02	8.59	3.84	9.32	10.52	6.84
ETFs & index	0.88	0.59	0.78	0.27	0.92	NA	1.11	0.62	0.78
Others	12.30	12.53	8.60	13.96	14.34	9.81	11.68	11.92	8.17
Foreign investors	4.25	5.81	3.05	1.09	3.49	1.23	5.43	6.58	3.67
Market wide (equal wt)	6.04	6.09	5.81	6.09	6.10	6.02	6.08	6.15	5.79
Market wide (value wt)	6.42	6.43	6.12	5.75	5.76	5.63	6.74	6.75	6.31

Notes: Elasticities are constructed from the nested model and aggregated as described in the paper. "NA" indicates missing/not applicable cases.

My interpretation.

- **Main ranking:** mutual funds are most elastic (≈ 14); life insurers and ETFs are extremely inelastic (≈ 1 and ≈ 0.9).
- **Market-wide elasticity:** around 6 for the average bond, implying nontrivial price impact from nonfundamental demand shocks.
- **IG vs HY:** IG is usually more elastic than HY at the type level (especially for mutual funds), consistent with more substitutability and/or easier trading in IG.
- **Time variation:** life insurers become even less elastic post-crisis (0.42 vs 2.69), while mutual funds/ETFs slightly increase. Aggregate market-wide elasticity changes modestly because composition and within-type elasticities move in opposite directions.

4.8 Table 8: Counterfactual equilibrium—credit quality migration (segmented markets)

Table 8: Counterfactual equilibrium: Credit quality migration in segmented markets (Table 8)

	Baseline		Low segmentation ($\lambda = 0.95$ for shocked group)		|Difference|	
	IG	HY	IG	HY	IG	HY
Panel A: IG downgrade (downgrade all IG bonds by one notch)						
Bonds predominantly held by life insurers						
< 5y	37.8	-31.4	38.8	-43.8	1.0	12.4
5y–10y	16.9	-12.4	17.4	-18.0	0.4	5.6
> 10y	6.9	-5.4	7.1	-7.9	0.2	2.5
Full sample	18.4	-19.2	18.9	-26.7	0.6	7.4
Financial crisis	23.5	-26.3	24.3	-39.3	0.8	13.0
Bonds predominantly held by mutual funds						
< 5y	33.2	-24.2	34.2	-35.3	1.0	11.1
5y–10y	14.5	-12.8	14.9	-18.9	0.4	6.1
> 10y	5.2	-5.7	5.4	-8.4	0.1	2.7
Full sample	17.0	-16.6	17.6	-23.0	0.5	6.4
Financial crisis	16.6	-22.9	17.2	-33.5	0.6	10.7
Panel B: HY downgrade (downgrade all HY bonds by one notch)						
Bonds predominantly held by life insurers						
< 5y	-1.2	41.1	-2.5	56.4	1.3	15.3
5y–10y	-0.5	17.0	-1.1	24.3	0.6	7.3
> 10y	-0.2	7.4	-0.5	10.3	0.3	2.9
Full sample	-0.6	26.0	-1.3	36.0	0.7	9.9
Financial crisis	-0.9	26.0	-1.8	45.6	0.9	19.6
Bonds predominantly held by mutual funds						
< 5y	-1.4	31.8	-2.41	40.95	1.1	9.2
5y–10y	-0.5	17.4	-0.9	22.1	0.4	4.7
> 10y	-0.2	7.1	-0.4	9.5	0.2	2.4
Full sample	-1.0	23.1	-1.8	29.9	0.8	6.9
Financial crisis	-0.6	23.5	-1.1	32.2	0.5	8.7

Notes: Entries are median changes in credit spreads (bps) vs actual. "Baseline" uses estimated λ. "Low segmentation" sets the λ entries for the downgraded rating group to 0.95.

My interpretation.

- **Direct effect:** downgrading a group increases its spreads (IG +18 bps overall; HY +26 bps overall in baseline). This is the straightforward risk/dislike-of-credit-deterioration effect.
- **Spillovers across groups:** the other group becomes relatively attractive $\Rightarrow$ spreads fall there (e.g., IG downgrade leads HY spreads to fall by about 19 bps in baseline).
- **Asymmetry:** spillover IG$\to$HY is much larger than HY$\to$IG (IG market is much larger, so rebalancing out of IG is huge relative to HY market size).

- **Role of segmentation:** making markets *less* segmented (high λ) amplifies both direct and spillover effects, especially for HY (because HY depends more on cross-group flows).
- **Maturity pattern:** short maturity bonds show larger effects (more exposed to the marginal investors with strong preferences, e.g., mutual funds).

4.9 Table 9: Counterfactual equilibrium—bond market fragility (outflows + who absorbs)

Table 9: Counterfactual equilibrium: Bond market fragility (Table 9)

	Absorb: Mutual funds		Absorb: ETFs/Index		Absorb: Life insurers	
	IG	HY	IG	HY	IG	HY
Panel A1: Bonds of institutions facing the outflows — outflows from mutual funds						
$<$ 5y maturities	19.8	25.0	19.6	42.5	25.2	50.2
5–10y maturities	7.8	3.9	7.7	17.8	8.5	19.9
$>$ 10y maturities	2.6	0.8	2.2	5.7	2.3	6.1
Full sample	13.7	25.0	14.7	41.3	17.2	48.2
Panel A2: Bonds of institutions facing the outflows — outflows from ETFs/Index						
	Absorb: ETFs/Index		Absorb: Mutual funds		Absorb: Life insurers	
$<$ 5y maturities	3.7	NA	8.1	NA	7.7	NA
5–10y maturities	2.2	NA	3.3	NA	2.8	NA
$>$ 10y maturities	0.4	NA	1.0	NA	0.7	NA
Full sample	3.2	NA	6.8	NA	6.2	NA

	Absorb: ETFs/Index		Absorb: Life insurers	
	IG	HY	IG	HY
Panel B1: Bonds of institutions absorbing the outflows — outflows from mutual funds				
$<$ 5y maturities	-5.9	NA	-5.8	NA
5–10y maturities	-2.2	NA	-2.2	NA
$>$ 10y maturities	-0.8	NA	-1.5	NA
Full sample	-4.7	NA	-2.8	NA
Panel B2: Bonds of institutions absorbing the outflows — outflows from ETFs/Index				
	Absorb: Mutual funds		Absorb: Life insurers	
$<$ 5y maturities	-2.7	-26.6	-1.6	NA
5–10y maturities	-1.2	-14.5	-0.5	NA
$>$ 10y maturities	-0.7	-7.5	-0.9	NA
Full sample	-2.2	-20.7	-0.9	NA

Notes: Entries are median changes in credit spreads (bps). Panel A: bonds predominantly held by the institutions facing outflows. Panel B: bonds predominantly held by institutions absorbing outflows. “NA” reflects negligible HY footprint for some types (e.g., ETFs/life insurers).

My interpretation.

- **Outflows from mutual funds are much more disruptive than outflows from ETFs:** spread increases are much larger, especially in HY and short maturities.

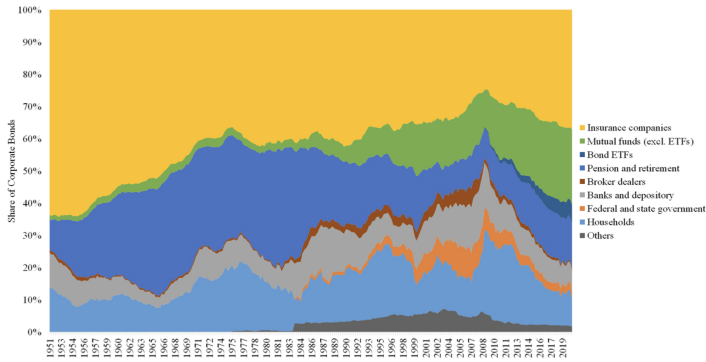


Figure 1
Institutional share of corporate bonds
The chart illustrates the share of institutional investors for the corporate bond market for the period 1951:4-2020:3. The data are quarterly from the U.S. Federal Flow of Funds account. We compute institutional shares excluding foreign owners of corporate bonds.

- **Who absorbs matters:** if other mutual funds absorb mutual-fund outflows, spread increases are smaller than if life insurers absorb them (preferences are more aligned within-type).
- HY effects explode when insurers must absorb HY sales (because insurers typically do not want HY; they require large compensation).
- **Panel B:** the institutions absorbing outflows receive inflows ⇒ spreads on their "home" bonds fall (negative changes), but effects are smaller in magnitude than the dislocations in Panel A.

5 Figures

5.1 Figure 1: Institutional share of corporate bonds (Flow of Funds, long-run)

What the figure shows. A stacked-area chart of the share of corporate bonds held by institutional sectors (1951Q4–2020Q3), excluding foreign owners.

My interpretation.

- The **ownership structure shifts massively over time**: insurers used to be much more dominant; mutual funds/ETFs rise strongly post-crisis.
- This motivates the paper's central question: if investor types have different demand elasticities and mandates, these composition shifts should move equilibrium spreads.

5.2 Figure 2: Evolution of within rating-group demand (life insurers vs mutual funds)

What the figure shows. Quarter-by-quarter coefficients for life insurers vs mutual funds, separately for IG and HY, for yield, maturity, bid-ask, bond size, and rating.

My interpretation.

- The **differences are persistent**: mutual funds are consistently more yield-elastic (especially in IG).

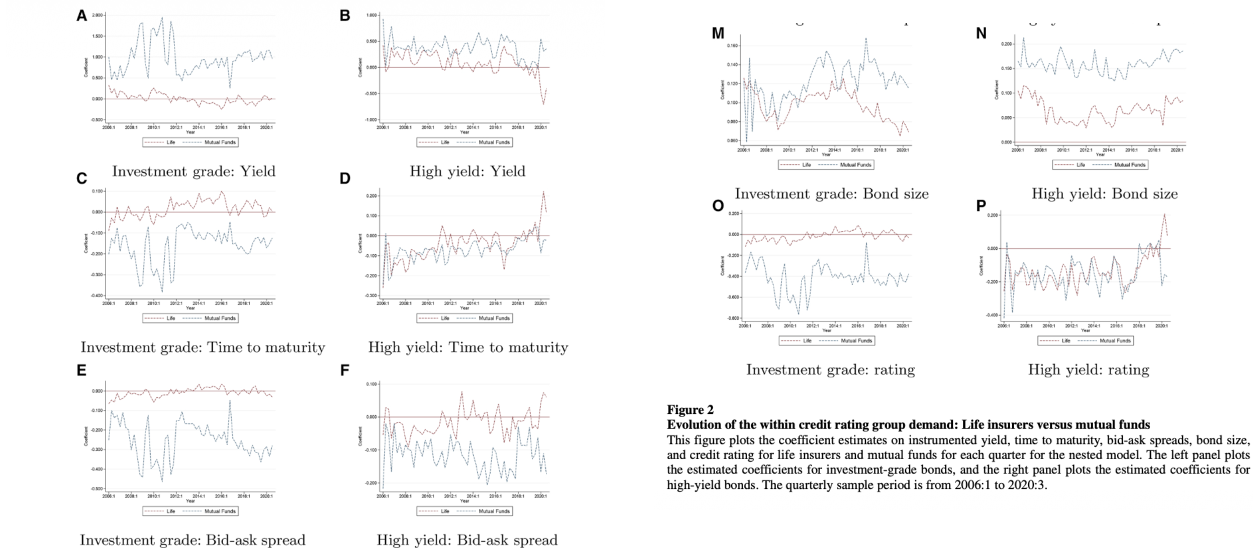


Figure 2
Evolution of the within credit rating group demand: Life insurers versus mutual funds
This figure plots the coefficient estimates on instrumented yield, time to maturity, bid-ask spreads, bond size, and credit rating for life insurers and mutual funds for each quarter for the nested model. The left panel plots the estimated coefficients for investment-grade bonds, and the right panel plots the estimated coefficients for high-yield bonds. The quarterly sample period is from 2006:1 to 2020:3.

- The **maturity preference wedge** is mainly an IG phenomenon: insurers tilt long, mutual funds tilt short.
- Mutual funds consistently dislike illiquidity (negative bid-ask coefficient), consistent with liquidity management due to redemption risk.
- Within IG, life insurers' rating coefficient is near zero (they mostly care about staying in IG), while mutual funds tilt toward safer bonds within the same broad bucket.

5.3 Figure 3: Cross-sectional dispersion of price impact (example quarter 2008Q4)

What the figure shows. Histograms of price impact for IG and HY bonds under: (A,B) baseline heterogeneity; (C,D) removing within-institution-type heterogeneity in elasticities; (E,F) removing within- and across-institution-type heterogeneity (only rating-group means left).

My interpretation.

- **Average price impact is large**, especially for HY (baseline shows much larger declines in HY prices for the same demand shock).
- **Dispersion is wide** in baseline (some bonds are much more fragile than others).
- Turning off within-type heterogeneity barely changes dispersion ⇒ the main driver is **differences across institution types** (who holds the bond), not fine heterogeneity within a type.
- Turning off across-type heterogeneity collapses dispersion ⇒ holdings by different investor types is the big source of cross-sectional price impact variation.

5.4 Figure 4: Counterfactual—undoing the rise of bond mutual funds and ETFs

What the figure does. Holds MF+ETF sector market share fixed at 2006Q1 levels by redistributing AUM to other investors; plots counterfactual spread changes over time, distinguishing high vs low MF/ETF ownership and by rating/maturity.

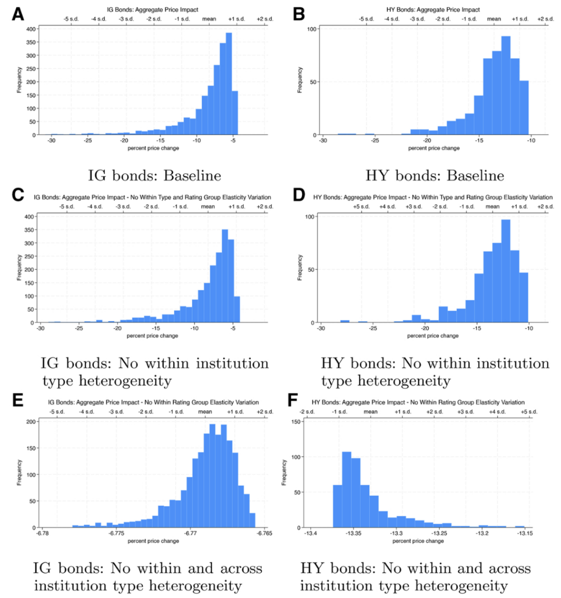


Figure 3
Cross-sectional dispersion of price impact
This figure plots the cross-sectional distribution of aggregate price impact for investment-grade and high-yield bonds for 2008:4. Panels A and B plot the distribution of aggregate price impact in the baseline case. Panels C and D plot the distribution of aggregate price impact when we do not allow for variation in elasticities within an institution type for a given rating group. Finally, panels E and F plot the distribution of aggregate price impact when we do not allow for variation in elasticities both within and across institution types for a given rating group.

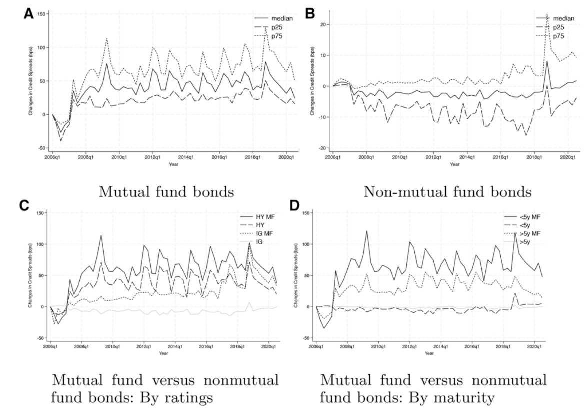


Figure 4
Counterfactual equilibrium: Undoing the rise of bond mutual funds and ETFs
This figure reports the counterfactual changes in credit spreads when we undo the rise of bond mutual funds and ETFs. That is, we restrict their relative market share to stay constant at the observed level at the beginning of our sample, 2006:1. We do so by redistributing the AUM of mutual funds and ETFs proportionally to all other investors. Panel A (panel B) documents the counterfactual changes in credit spreads for bonds with either high (low) mutual fund or ETF ownership. Panels C and D report corresponding results comparing investment-grade to high-yield bonds and short (<5 years) to long maturity bonds (>5 years). The sample period is quarterly from 2006:1 to 2020:3.

My interpretation.

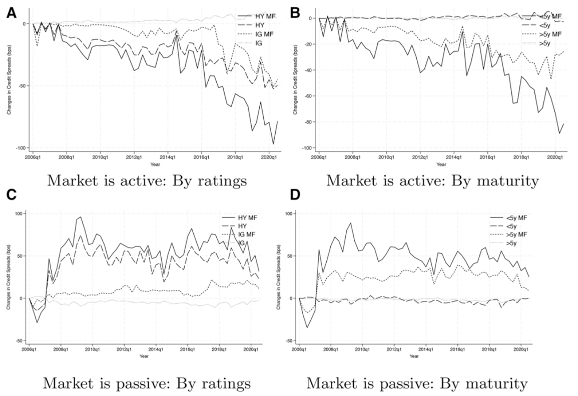
- Bonds with **high mutual fund ownership** would have **much higher spreads** in the counterfactual (mutual funds being large marginal buyers compress spreads in reality).
- Effects are **largest in crises** and for **HY and short-maturity bonds** (these are the bonds mutual funds like more than the alternative holders, and/or where alternative holders are less elastic).
- Bonds with low MF ownership can see small or even negative spread changes because other investors receive inflows and bid up their preferred bonds.

5.5 Figure 5: Counterfactual—active versus passive bond funds

What the figure does. Two scenarios: (1) ETF growth reallocated to mutual funds (market is "more active"); (2) mutual fund growth reallocated to ETFs (market is "more passive"). Spread changes shown by rating/maturity and by high vs low MF ownership.

My interpretation.

- If passive growth had instead been active (scenario 1), spreads fall more for bonds mutual funds like (HY, short maturity), and the effect grows over time because ETFs grew a lot in reality.
- If active growth had instead been passive (scenario 2), spreads rise for riskier/shorter bonds because ETF demand is more benchmark/IG-oriented and less tilted to HY/short.
- Net message I take: **the composition of the asset-management sector (active vs passive) changes equilibrium credit spreads**, not just total AUM.



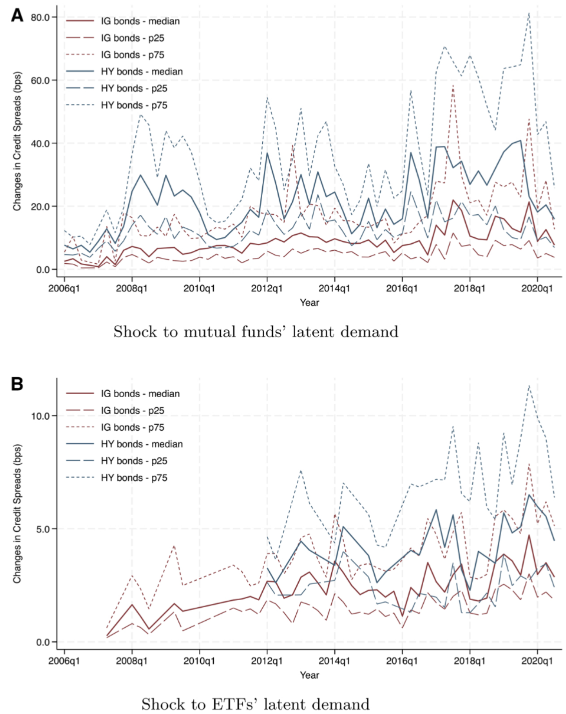
Market is active: By ratings

Market is active: By maturity

Market is passive: By ratings

Market is passive: By maturity

Figure 5
Counterfactual equilibrium: active versus passive bond funds
This figure reports the counterfactual changes in credit spreads when we redistribute AUM from passive (ETFs and indexers) to mutual funds and vice versa. That is, in panels A and B (panels C and D) we restrict the relative market share of passive (active) funds to stay constant at the observed level in 2006:1 by redistributing AUM from passive (active) funds to active (passive) funds. We report the results comparing investment-grade to high-yield bonds and short (<5 years) to long maturity bonds (>5 years). Moreover, we distinguish bonds with high and low mutual fund ownership (MF). The quarterly sample period is from 2006:1 to 2020:3.



Shock to mutual funds' latent demand

Shock to ETFs' latent demand

Figure 6
Bond market fragility: Price impact of demand shocks
This figure reports the counterfactual changes in credit spreads when either all mutual funds (panel A) or all ETFs (panel B) experience a 25% latent demand shock. Panels A and B document the cross-sectional distribution of the credit spread changes both for investment-grade and high-yield bonds by reporting the median, the 25th percentile, and the 75th percentile of the distribution. The quarterly sample period is from 2006:1 to 2020:3.

5.6 Figure 6: Bond market fragility—price impact of demand shocks (25% latent demand shock)

What the figure shows. Time series of spread changes (median, 25th, 75th percentiles) for bonds with high mutual fund share (Panel A) versus high ETF share (Panel B), under a 25% latent demand shock to that sector.

My interpretation.

- Shocks to **mutual funds** create **larger spread spikes** than comparable proportional shocks to ETFs.
- HY bonds are consistently more fragile (bigger spread responses).
- Even when ETF sector grows, ETF shocks remain more muted because ETFs hold mostly IG and large issues that have broader ownership by other (more elastic) investors.

6 Conclusion

6.1 What the paper ultimately concludes

- The corporate bond market is **not well described by a representative-investor view** where "prices move only with fundamentals" and arbitrage is effectively perfectly elastic. Instead, **equilibrium credit spreads depend on who holds the bonds**, because major institutional sectors have **different mandates, preferences, and demand elasticities**.

- A key state variable for corporate bond pricing is the **composition of institutional demand** (the cross-sectional distribution of AUM across investor types and which bonds they hold).
- Empirically, the market exhibits **strong segmentation across broad rating groups** (investment grade vs. high yield), which implies that shocks in one segment do not seamlessly get arbitraged away by reallocations across the other segment.
- Using the estimated model, the paper concludes that recent **ownership trends** have **systematic effects on equilibrium spreads and corporate funding costs**:
  - The **rise of bond mutual funds** has **lowered spreads**, especially for **short maturity** and **high-yield** bonds.
  - The **rise of ETFs/indexers** (benchmark-oriented, mostly IG) can **increase the relative borrowing costs of riskier issuers** by shifting demand away from riskier bonds.
- The paper also concludes that **bond-market fragility depends crucially on (i) who faces outflows, (ii) who provides liquidity, and (iii) what bonds are being sold and whether they are in the buyers' investment universes**. In particular, comparable outflows from mutual funds generate larger dislocations than outflows from ETFs.

6.2 Summary

The paper builds and estimates a **demand-based equilibrium model of corporate bond pricing** that directly incorporates **heterogeneous institutional investors**. The starting point is a set of empirical facts from holdings data: (i) insurance companies, mutual funds, and ETFs dominate corporate bond ownership; (ii) these sectors have very different mandates (duration-matching for insurers, redemption-liquidity for mutual funds, benchmarking for ETFs); and (iii) portfolios are strongly segmented along the **investment-grade (IG)** vs. **high-yield (HY)** boundary.

To capture these facts, the paper estimates a **nested demand system**: within each rating group (IG or HY), institutions choose among individual bonds based on yields and characteristics (maturity/duration, credit risk, liquidity, issuance size), and across rating groups they choose how much IG vs. HY to hold using an "inclusive value" (a summary of how attractive the whole group is). Market clearing then pins down equilibrium bond prices and spreads. A key identification challenge (yields are endogenous to demand) is handled via a mandate-based instrument derived from persistent investment universes.

With the estimated demand system in hand, the paper delivers (i) **elasticity estimates** by institution type and rating segment, (ii) **segmentation estimates** (imperfect substitution across IG and HY), (iii) **bond-level price impact/fragility** implied by demand shocks, and (iv) **counterfactual equilibria** that quantify how spreads would change under ownership shifts (growth of mutual funds vs. ETFs), under downgrade waves, and under large redemptions with different liquidity providers.

6.3 Why the mechanism works

Core mechanism: "who is marginal" determines spreads. In equilibrium, each bond's price adjusts until the **marginal dollars of demand** are willing to hold the outstanding supply. If the investor base is dominated by **inelastic** holders (e.g., buy-and-hold insurers, benchmarked ETFs), then clearing the market after a shock requires **large price changes** (large spread moves).

If the investor base has more **elastic** holders (e.g., mutual funds willing/able to substitute and trade), then the same shock is absorbed with **smaller price changes**.

Why segmentation matters. Even if there are some elastic investors, **segmentation across IG and HY** means their willingness/ability to shift dollars across the IG–HY boundary is limited (mandates, regulation, benchmarks). Therefore:

- Shocks can be **localized**: HY can become severely dislocated if few investors can step in.
- Spillovers are **asymmetric**: because IG is much larger, reallocations out of IG can move HY a lot (as a fraction of HY market), whereas HY reallocations barely move IG.

Why ownership trends matter. When mutual funds grow, more marginal demand comes from an investor type that is (estimated to be) **more elastic** and also **prefers HY and short maturity more than insurers**. That shifts equilibrium toward **lower spreads** precisely in those segments. When ETF/indexing grows at the expense of active mutual funds, demand becomes more **benchmark/IG tilted**, which can raise **relative** spreads for riskier issuers.

6.4 The paper's punchline in "numbers"

These are the quantitative statements I treat as the paper's headline conclusions:

- **Market-wide demand elasticity** for the average corporate bond is around ≈ 6, far from the "nearly infinite" elasticity that standard frictionless models would suggest.
- **Sector heterogeneity is large:**
  - Mutual funds are the most elastic (about ≈ 14 on average).
  - Life insurers and ETFs/indexers are very inelastic (about ≈ 1 and ≈ 0.9, respectively).
- **IG tends to be more elastic than HY within most sectors** (e.g., for mutual funds, IG elasticities are much larger than HY elasticities), consistent with either fewer close substitutes / higher trading frictions in HY.
- **Across-group substitution is low:** the nested substitution parameter is around ≈ 0.2, consistent with meaningful IG–HY segmentation.
- **Price impact is economically large and heterogeneous**, especially in HY; and most cross-sectional variation in fragility is driven by **differences across investor types** holding the bond (not fine differences within a type).

6.5 Broader implications

- For empirical asset pricing: corporate credit spreads are shaped by **demand composition** in addition to cash-flow/default fundamentals and discount-rate risk.
- For policy: regulations or interventions that shift AUM or constraints across insurers vs. mutual funds vs. ETFs can move credit spreads and hence **corporate financing costs**.
- For stress/fragility: "fire sale" risk is not only about the size of outflows, but about **who sells**, **who can buy**, and **how segmented the relevant segment is**.